



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Understand Percent

Lesson 4-1 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

Language Objective: I can explain what a percent means using the words per hundred, part, whole, and equivalent ratio.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Understand Percent

Percent

Per hundred

Decimal grid

Part-to-whole

Greater than 100%



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A ratio comparing a number to 100 is a .

2

The word that means “per hundred” is .

3

The meaning hidden inside the word percent is

.

4

A 10-by-10 grid of 100 squares used to show a percent is a

.

5

A comparison of one part against the whole amount is a

comparison.

- 6 A percent above 100 means the amount is than one whole.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What percent of the gray kitten's weight does the orange kitten weigh, and how does the tape diagram show it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Understand Percent** — A percent is a ratio that compares a number to 100, so every percent answers the question “how many out of 100?”

Comprender el porcentaje — Un porcentaje es una razón que compara un número con 100, así que todo porcentaje responde “¿cuántos de cada 100?”

- ☐ **Percent** — A ratio that compares a number to 100. The word means “per hundred,” and the symbol is %.

Porcentaje — Una razón que compara un número con 100. La palabra significa “por cada cien” y su símbolo es %.

- ☐ **Per hundred** — The meaning hidden inside the word percent: for every group of 100. It is why every percent can be written as a fraction over 100.

Por cada cien — El significado dentro de la palabra porcentaje: por cada grupo de 100. Por eso todo porcentaje se puede escribir como fracción sobre 100.

- ☐ **Decimal grid** — A 10-by-10 grid of 100 squares. Shading squares shows a percent directly, one square per percent.

Cuadrícula decimal — Una cuadrícula de 10 por 10 con 100 cuadros. Sombrear cuadros muestra un porcentaje directamente, un cuadro por cada uno por ciento.

- ☐ **Part-to-whole** — A comparison of one part against the whole amount. In a percent, the whole is always 100%.

Parte a todo — Una comparación de una parte con el total. En un porcentaje, el total siempre es 100 %.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The gray kitten is the ____, so its weight is ____%. The orange kitten needs one whole tape plus ____ of another. Adding the parts, ____% + 50% = ____%. A percent greater than 100% means the orange kitten weighs ____ than one whole gray kitten.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 150% is more than one whole, and 0.5% is a tiny piece smaller than 1%.

Evidence: Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

Reasoning: This evidence connects the definition of understand percent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The gray kitten is the ____, so its weight is ____%. The orange kitten needs one whole tape plus ____ of another. Adding the parts, ____% + 50% = ____%. A percent greater than 100% means the orange kitten weighs ____ than one whole gray kitten.

- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Understand Percent
2. **Notes 2:** Percent
3. **Notes 3:** Per hundred
4. **Notes 4:** Decimal grid
5. **Notes 5:** Part-to-whole
6. **Notes 6:** Greater than 100%
7. **Watch for this mistake:** *A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor — $10 \times 10 = 100$, so $7 \times 10 = 70$, giving 70%. Reading it as 7% describes seats that are nearly empty when they are nearly full. A second mistake is treating a percent as an amount: 150% of a 200-point goal is $1.5 \times 200 = 300$ points, not 150 points.*
8. **Try It 1:** A. Twice as much as you need — 100% is what you need, so 200% is two of those — twice as much.
9. **Try It 2:** — Dividing a percent by 100 tells you how many wholes it is.